

Nitin Raj

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SUMMARY

Second-year Information Technology student at Narula Institute of Technology with 3 years of hands-on experience as a Backend and React Developer. Specialized in designing, building, and shipping production-ready applications across modern JavaScript ecosystems. Strong expertise in React, React Native, Node.js, Express.js, Socket.IO, MongoDB, and Firebase. Proven track record in developing scalable full-stack web platforms, cross-platform mobile applications, and real-time systems. **Currently advancing expertise in Cybersecurity, Aerospace software engineering**, Java, Next.js, and cloud-native architectures.

TECHNICAL SKILLS

Languages & Frameworks: JavaScript (ES6+), TypeScript, React.js, React Native, Node.js, Express.js, Next.js, HTML5, CSS3, Java, Python

Databases & Tools: MongoDB, Firebase, Git, GitHub, RESTful APIs, Socket.IO, WebSockets, Postman, Authentication

Platforms & DevOps: Linux (Arch), AWS, Docker, Vercel, Render, Git Version Control, LaTeX, VS Code, npm/yarn, PWA, SSR

PROJECTS

OrbitShield (Orbital Risk Engine) | *Python, FastAPI, SQLAlchemy, Skyfield/SGP4, React, Three.js*

- Engineered a real-time satellite conjunction detection system tracking 16,000+ live satellites using SGP4 orbital propagation, reducing collision-pair comparisons from ~128M (naive) to a few hundred per group via a two-stage spatial filter (altitude + orbital-plane banding)
- Built a 6-endpoint REST API with an automated background scheduler for continuous data refresh, plus a 3D interactive globe (React + Three.js) visualizing live satellite positions and orbital paths

WorkspaceDev | *React.js, Node.js, MongoDB, Express.js, Socket.IO*

- Engineered collaborative project management platform with real-time WebSocket updates, organization-based workspace architecture, and role-based access control for team coordination
- Integrated real-time notifications and activity feeds using Socket.IO, improving team communication efficiency by 40%

CineCue - Movie Discovery App | *React.js, TMDB API, Axios, CSS3*

- Developed comprehensive movie discovery application leveraging TMDB API with advanced search, filtering by genre/year/popularity, and responsive design across 10,000+ entries
- Optimized API request handling with debouncing techniques, reducing unnecessary API calls by 60%

Real-Time GPS Tracking System | *Node.js, Socket.IO, Leaflet.js, Express.js*

- Built production-ready live location tracking system streaming geographic coordinates in real-time using WebSocket technology with Leaflet.js interactive map visualization
- Architected scalable WebSocket server handling 100+ concurrent connections with sub-second latency

EDUCATION

Narula Institute of Technology

Sep. 2025 – Present

Bachelor of Technology in Information Technology

Kolkata, West Bengal

- Relevant Coursework:** Data Structures & Algorithms, OOP, DBMS, Web Technologies, Computer Networks
- Technical Focus:** Full-stack development, mobile applications, real-time systems, cybersecurity